Alberto Reyes Castro

Applied AI Engineer | Multimodal & On-Device ML | MSc AI Technology

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Availability: open to suitable part-time AI/ML roles now; full-time during official university vacation periods and for the Advanced Practice placement (Sep 2027 – Jan 2028); graduate roles from early 2028.

Right to work: Student visa valid to May 2028 (up to 20 hours/week during term; full-time outside term-time and on an approved course placement); eligible to apply for the 18-month Graduate visa after completing the MSc.

Summary

Applied AI engineer and physics graduate completing an MSc Artificial Intelligence Technology with Advanced Practice at Northumbria University London, with Distinction-level first-semester module results. At TeachScribe, designed and built an on-device multimodal photo-retrieval system now live in production and shipped to 50+ UK schools, spanning model selection, INT8 ONNX deployment, native Android preprocessing and physical-device evaluation. Independent work spans multi-agent orchestration and reliability engineering, ModelHawk ML auditing and public multimodal/speech-inference tooling. Open to Applied AI and ML engineering opportunities across multimodal systems, efficient inference, AI platforms and reliable agentic systems.

Experience

Applied AI Engineer – On-Device Multimodal ML **Jul 2026 – Present**
TeachScribe – UK EdTech start-up in 50+ schools | DfE AI Tools for Education
Full-time Jul–Oct 2026; part-time alongside MSc thereafter

- Designed, built and evaluated the fully on-device multimodal photo-retrieval pipeline now live in production and shipped through TeachScribe to 50+ UK schools; written observations retrieve local classroom photos while inference stays on the tablet and the teacher confirms matches.
- Benchmarked five pretrained vision–language models across three structured retrieval evaluations, balancing Hit@1/Hit@5, model size, CPU cost, deployability and licensing; selected TinyCLIP 39M/16, which ranked the correct image first in 17/20 queries on the initial 16-candidate model-selection set and achieved higher Hit@1 than MobileCLIP-S0 across all three evaluated sets.
- Implemented ONNX Runtime inference with dynamic INT8 quantisation, native Kotlin preprocessing, reusable sessions and a RAM + SQLite embedding cache; the text encoder fell from 178.5 MB FP32 to 45.0 MB INT8 (**-74.77%**), with 0.978 mean cosine similarity to FP32 on real text queries.
- On a Huawei MediaPad M3 Lite 8 (Android 7), measured 108 consecutive cached searches over 44 photos at **441 ms median / 459 ms P95**; lazy image-model loading kept idle memory at 76.3 MiB, and the tested workload showed no observed throttling.

Public multimodal portfolio links: GitHub | Live demo

Python & Scientific Communication Intern **Mar 2025 – May 2025**
Instituto de Astrofísica de Canarias (IAC) – astrophysics research institute
Full-time internship | Tenerife, Spain

- Built Python workflows for multilingual scientific-interview transcription and segmented subtitling using the internship's original speech-processing stack (including `SpeechRecognition`), including material featuring Nobel Prize-winning physicists Donna Strickland and John Mather. After the placement, independently re-engineered the workflow into a reusable faster-whisper/CTranslate2 pipeline with hardware-aware CPU/GPU execution, FFmpeg preprocessing, bounded fallbacks and CI.
- Built a human-verified matching workflow comparing 67 outreach articles with 750+ scientific-publication titles using multilingual text normalisation, regular expressions and fuzzy matching.
- Wrote science-outreach content for the IAC and *Astronomía* magazine; completed the assessed 250-hour placement with a final mark of 9.2/10.

Selected Projects

Multi-Agent Orchestration & Reliability – Agent Control Center | Live demo

- Built a private Python/PowerShell orchestration system with capability-aware local/cloud routing, durable task state, account-scoped quota controls, execution leases and heartbeats, bounded recovery, separate execution/review roles, deterministic validation and human merge gates; published a sanitized interactive demonstration of the control-plane architecture.

ModelHawk – ML Notebook Audit Tool | Live demo

- Built and deployed a Next.js/TypeScript static-audit tool for Jupyter notebooks with deterministic checks for leakage, validation, metrics and reproducibility; Dockerised Render deployment and CI exercise the production container, audit API and LaTeX/PDF path end to end.

Adaptive Whisper Transcription Pipeline

- Developed a reusable faster-whisper/CTranslate2 workflow with CPU/CUDA-aware execution, FFmpeg preprocessing, fingerprinted caching, bounded fallback presets, quality checks and atomic TXT/SRT outputs, with deterministic CI across Python 3.10–3.12.

Education

MSc Artificial Intelligence Technology with Advanced Practice **Jan 2026 – Jan 2028**
Northumbria University, London Campus *London, UK*
First-semester performance at **Distinction level**, including module results of **85/100, 75/100 and 70/100**. Current study includes *AI Agents Unleashed*; previous modules include *Machine Learning on Cloud*, *Programming for AI*, *AI for IoT* and *Contemporary Computing and Digital Technologies*.

BSc in Physics **Sep 2018 – Jul 2025**
Universidad de La Laguna *Tenerife, Spain*
Final grade: **7.13/10 (approx. UK 2:1)**; 240 ECTS, with training spanning mathematics, scientific computing, numerical methods, signal/image processing, laboratory analysis and data-driven problem solving.
Selected module results: General Relativity 10/10; Signals & Images 9.9/10; Differential Calculus 9.5/10; Electronic Physics 9.0/10; Statistical Physics 8.7/10.
Final Degree Project (**9.4/10**), written and defended in English: *Structural Mechanisms of Ferroelectricity and Ferroelasticity in* β'-$RE_2(MoO_4)_3$. My supervisor proposed developing the work into a manuscript for submission to a high-impact peer-reviewed journal; manuscript preparation and further polishing are ongoing.
Awarded the **University of La Laguna 2025 Academic Merit Convocation**.

Technical Skills

Programming and data: Python, TypeScript/JavaScript, Kotlin, SQL/SQLite, PowerShell, Bash, C++/Arduino, R, pandas, NumPy, SciPy, Matplotlib and Jupyter.
Machine learning and evaluation: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, TensorFlow/Keras, model benchmarking, Hit@K retrieval evaluation, imbalanced learning, cross-validation, threshold selection and error analysis.
Multimodal and on-device AI: vision–language models (CLIP/TinyCLIP), embeddings, ONNX Runtime, INT8 quantisation, Android (React Native/Expo, Kotlin), native preprocessing, persistent embedding caches, latency/memory/energy profiling and physical-device validation.
LLMs and agent systems: multi-agent orchestration, capability-aware routing, durable task state, leases/heartbeats, bounded recovery, quota-aware provider controls, deterministic validation, separate review roles and human approval gates.
Engineering and cloud: Git/GitHub, GitHub Actions, Docker, Render, REST APIs, Next.js/React, Node.js, Linux, Windows/PowerShell and Google Cloud.
Speech, embedded AI and scientific computing: faster-whisper/CTranslate2, FFmpeg, voice-activity detection, ESP32-S3, Edge Impulse/TinyML, Wokwi, Blynk, regular expressions, fuzzy matching, numerical methods and LaTeX.

Open-Source Contribution

Vizro / Dash Rendering Regression Analysis

- Reproduced and isolated a Vizro/Dash tab-rendering regression through dependency comparisons and minimal pure-Dash tests, separating it from Plotly graph rendering; a Vizro maintainer acknowledged the investigation as useful in narrowing the defect and developing a fix.

Selected Publications & Scientific Communication

- *The Partial Solar Eclipse That Reminded Canary Islanders of the Value of Looking Skyward* (Spanish) – co-authored with Ariadna Mesa and Hugo Fernández; *Astronomía* magazine, No. 311 (May 2025).
- *The WEAVE spectrograph captures the interaction of supernova SN 2023ixf with its surroundings one year after its explosion* – English-language version written for IAC News (Apr 2025).
- *The Partial Solar Eclipse of March 2025* (Spanish) – authored for the IAC *Vía Láctea* science-outreach blog (Mar 2025).

Languages & Selected Training

English: C1 – IELTS Academic UKVI 7.0/9.0 and Cambridge C1 Advanced Certificate. **Spanish:** Native. **Japanese:** Intermediate study; preparing for JLPT N3. **French:** Limited working proficiency.
CS50's Introduction to Databases with SQL – Harvard University (in progress).
AWS Educate Machine Learning Foundations (2026).
Artificial Intelligence and Big Data Applied to the Healthcare Sector – 325-hour course (Canary Islands Employment Service, 2025): Python, pandas, scikit-learn, TensorFlow/Keras, Docker/APIs, cloud platforms, Big Data and R.

Activities

Competitive basketball with London Knights (2026 – present).